Getting Started with Geographic Information Systems (GIS)

Intro to GIS + Map Creation Basics

WORKSHOP

VANESSA CASILLAS · DANIEL PEREZ

What We'll Cover Today

Part 1 — This Session

- What GIS is and why it matters
- Basic concepts of spatial data
- How to create a simple map

Part 2 — Hands-On Session

- Building a map step-by-step
- Adding and styling data layers
- Exploring data and creating a final map product

What is GIS?

Geographic Information Systems (GIS) is a technology for capturing, storing, analyzing, and visualizing geographic data. In simple terms: **GIS connects data + location** to help us understand the world.

Capture

Collect geographic data from the real world

Store

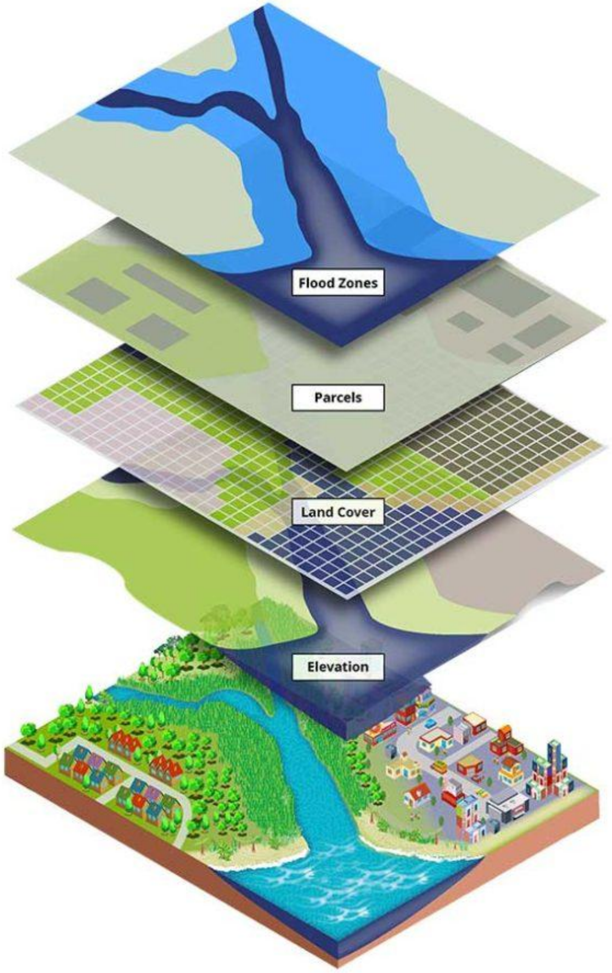
Manage and organize spatial information

Analyze

Discover patterns and relationships

Visualize

Display insights on interactive maps



What Makes GIS Powerful?

GIS helps answer critical questions — and is used across many fields.

Where are things?

Locate resources, hazards, or services precisely.

What patterns exist?

Reveal trends invisible in spreadsheets.

How do places relate?

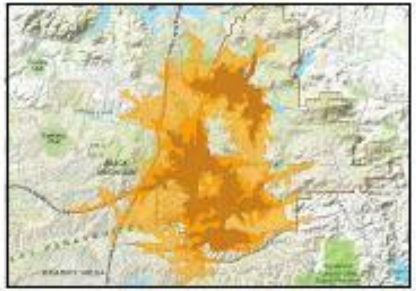
Understand spatial relationships between locations.

What has changed?

Track changes over time across a landscape.

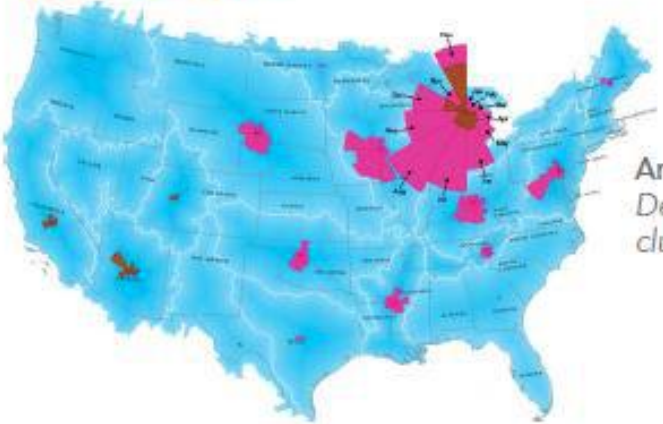
Used in: Public health · Disaster response · Transportation · Equity analysis · Campus planning

Understand Places



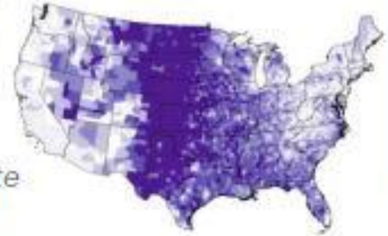
Analysis Tools: *Attribute queries, spatial queries; proximity analysis*

Detect Patterns



Analysis Tools: *Density analysis and cluster analysis*

Determine Relationships

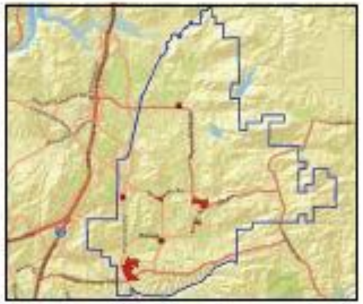


Analysis Tools: *Attribute joins, spatial joins, and overlay analysis*

Make Predictions

Analysis Tools: *Interpolation, regression, surface analysis*

Find Locations



Analysis Tools: *Site suitability, location-allocation, cost corridors*

https://www.esri.com/about/newsroom/insider/the-importance-of-where

Bus and Rail System Overview map

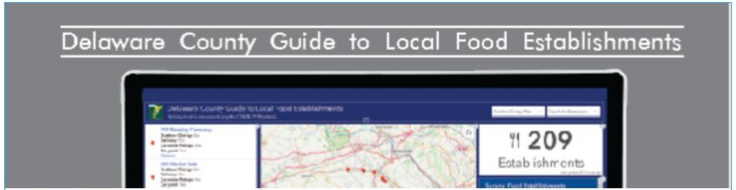
DCR Design LLC
by DCR Design

14 0

Nevada's Population Change: A Review of Population Growth and Decline in the State of Nevada

Legislative Counsel Bureau
by Hildi Cruz and Robin Evans

8 0



Delaware County Guide to Local Food Establishments

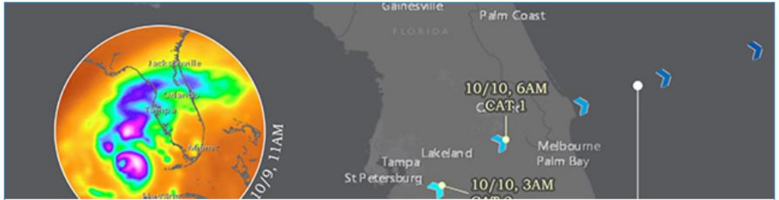
Delaware County Planning Department
by Amanda Hagan

10 1

https://mapgallery.esri.com/

Protected Areas Database of the United States (PAD-US) - IUCN Protected Area Categories

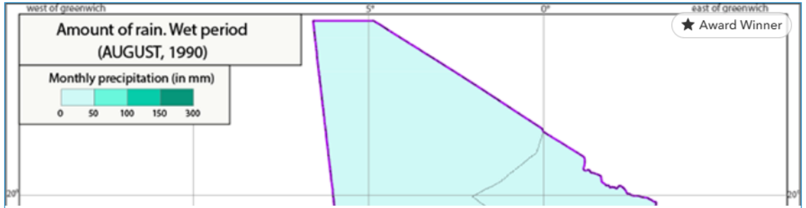
ey
cal Survey prepared i



Hurricane Milton Strength Timeline, 10/5/2024 - 10/10/2024

NASA
by Ariana L. McDermott, Lauren K. Hill-Beaton

5 0



Amount of Rain in Mali in Wet and Dry Periods

Africa Business and Consulting Mali
by Souleymane Sanogo

47 17

Willamette Valley American Viticultural Area - A Geologic History of Soils

Cascade GIS
by Erica McCormick and Rick Zeeb

7 0

Nipah Virus: Risk Analysis of Outbreak Potential in South and Southeast Asia

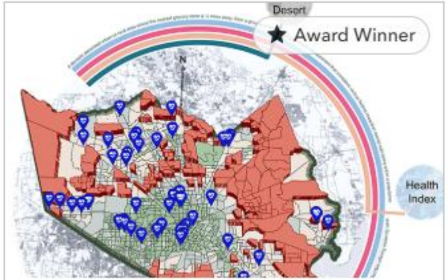
Tufts University
by Shannon O'Connor

10 0

Intelligent Supervision with Geospatial Technology

Osinergmin (Lima, null)
by ROGER LOPEZ TUESTA; LIZBETH TORRES…

578 0



Geography of Maternity Care Deserts

Harris County
by Meliadis Barbara, L. Venigalla, J. Wong, A.…

575 0

Listrik Desa (Lisdes) 2.0: Electrifying the Nation with Renewable Energy

PT PLN (Persero) Kantor Pusat
by Wahyu Ahadi Rouzi

190 0

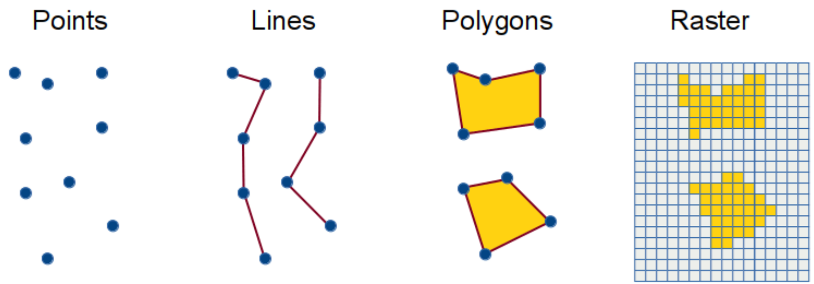
Ghana Cocoa Board

Ghana Cocoa Board
by Felix Owusu Anyimah

27 0

https://mapgallery.esri.com/

Core Idea: Data + Location



Every GIS dataset has two essential parts working together.

📍 Spatial Data — Where

- **Points** — trees, buildings, schools
- **Lines** — roads, rivers, pipelines
- **Polygons** — boundaries, campuses, zones

📋 Attribute Data — What

- **Name** — label or identifier
- **Type** — category or classification
- **Population** — numeric values
- **Status** — condition or state
- **Any description** — additional context

Key GIS Components

Basemap

The background reference layer — streets, terrain, or satellite imagery.

Data Layers

Your custom geographic information overlaid on the basemap.

Symbology

Colors, shapes, and sizes that represent different data values.

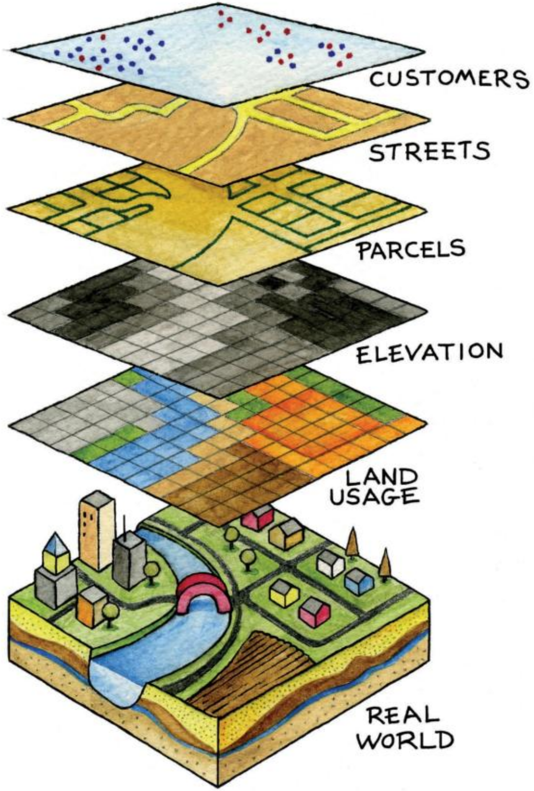
Pop-ups

Information windows that appear when you click a feature.

Legend

A guide explaining what each symbol and color means.

What is a Map in GIS?



A GIS map is **more than a static image**. It's a dynamic collection of **layers** — each holding a different dataset that can be turned on or off, styled, and analyzed.

> Think of it like a stack of transparent sheets. Each sheet represents a different dataset — roads, buildings, parks — layered together to tell a complete story.

How to Make a Map: Basic Workflow

01

Define Your Question

What are you trying to show? Who is the audience?

02

Find or Load Your Data

CSV files, GIS servers, open data portals, or institutional datasets.

03

Add a Basemap

Choose a background reference map.

04

Add Data Layers

Load your spatial data as points, lines, or polygons.

05

Style Your Layers

Apply colors, symbols, and sizes to represent your data.

06

Configure Pop-ups

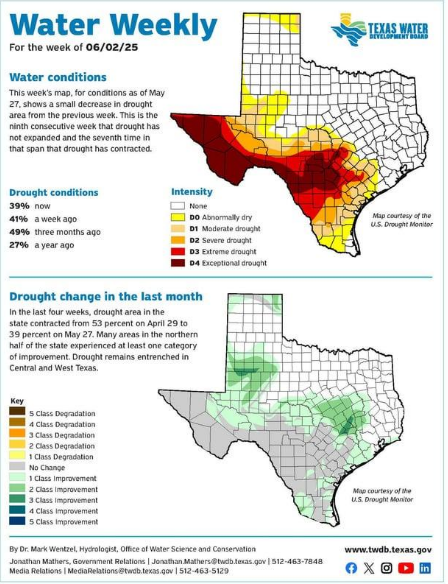
Set up information windows for map features.

07

Review and Interpret

Check your map and draw conclusions from what you see.

Step 1: Define Your Purpose



Water Weekly

For the week of **06/02/25**

TEXAS WATER DEVELOPMENT BOARD

Water conditions

This week's map, for conditions as of May 27, shows a small decrease in drought area from the previous week. This is the ninth consecutive week that drought has not expanded and the seventh time in that span that drought has contracted.

Drought conditions

39% now
41% a week ago
49% three months ago
27% a year ago

Intensity

None
D0 Abnormally dry
D1 Moderate drought
D2 Severe drought
D3 Extreme drought
D4 Exceptional drought

Map courtesy of the U.S. Drought Monitor

Drought change in the last month

In the last four weeks, drought area in the state contracted from 53 percent on April 29 to 39 percent on May 27. Many areas in the northern half of the state experienced at least one category of improvement. Drought remains entrenched in Central and West Texas.

Key

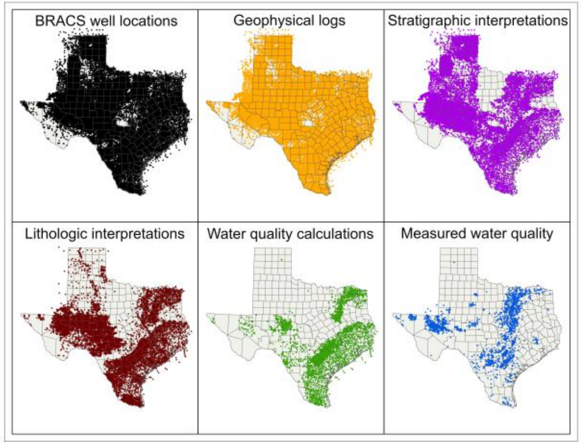
5 Class Degradation
4 Class Degradation
3 Class Degradation
2 Class Degradation
1 Class Degradation
No Change
1 Class Improvement
2 Class Improvement
3 Class Improvement
4 Class Improvement
5 Class Improvement

Map courtesy of the U.S. Drought Monitor

By Dr. Mark Wentzel, Hydrologist, Office of Water Science and Conservation

Jonathan Mathers, Government Relations | Jonathan.Mathers@twdb.texas.gov | 512-463-7848
Media Relations | MediaRelations@twdb.texas.gov | 512-463-5129

www.twdb.texas.gov



Before building any map, start with a clear purpose. Ask yourself:

What am I trying to show?

Define the core message or insight.

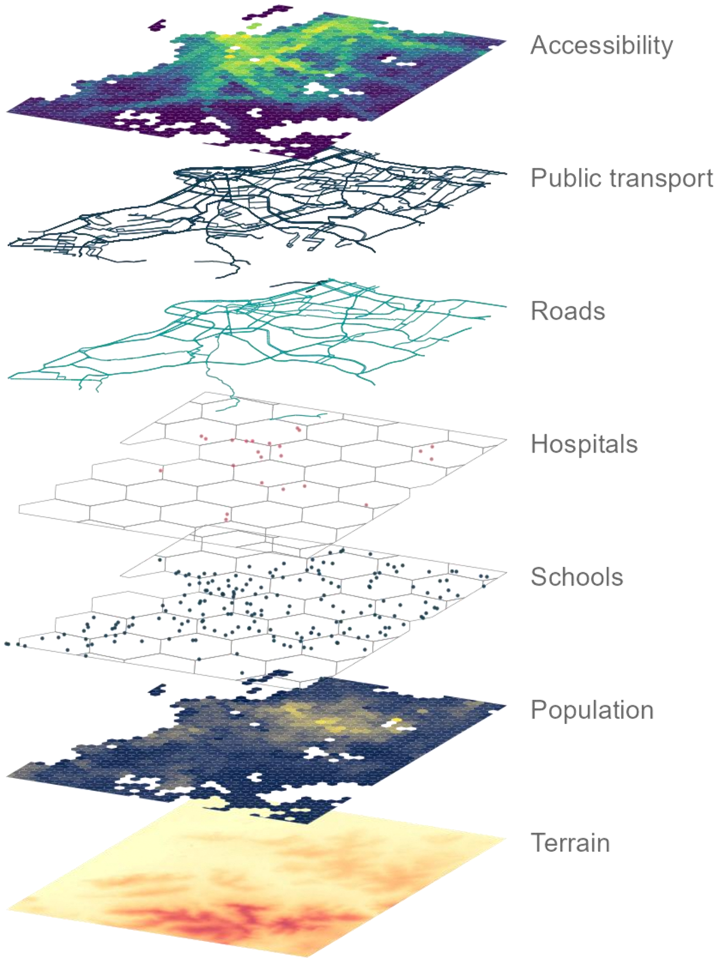
Who is the audience?

Tailor complexity and design to the viewer.

What decision will this support?

Maps should drive action or understanding.

Example questions: Where are campus resources? How are services distributed? Where are environmental risks highest?



Step 2: Add Your Data

Data is the foundation of every GIS map. You can bring data in from many sources.

Spreadsheets (CSV)

Tables with addresses or coordinates

GIS Servers

Live data feeds from institutional systems

Open Data Portals

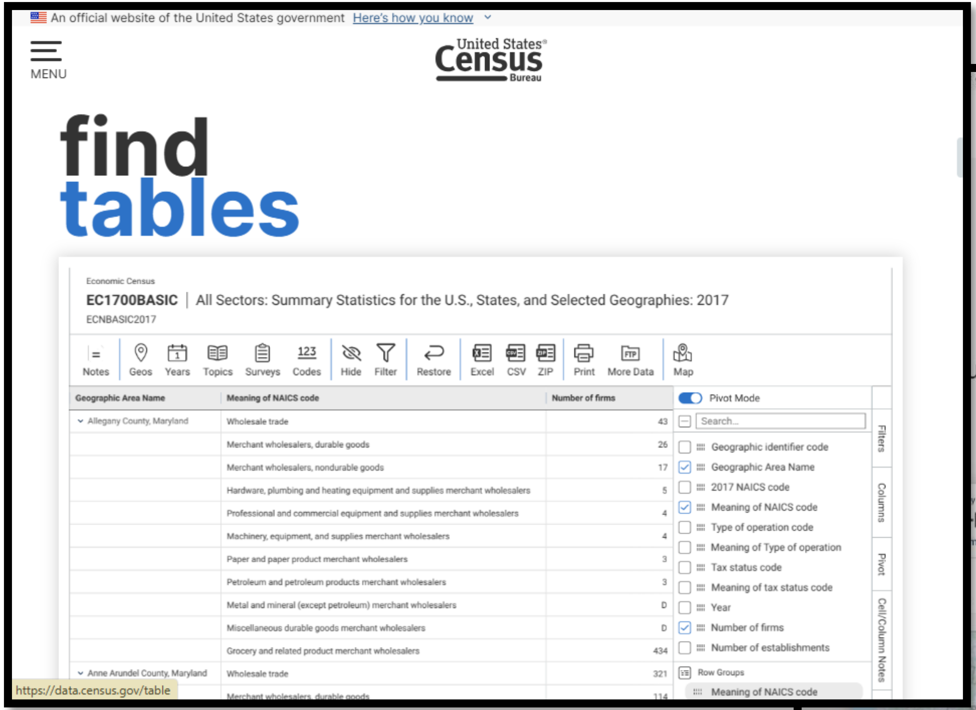
Public datasets from governments and organizations

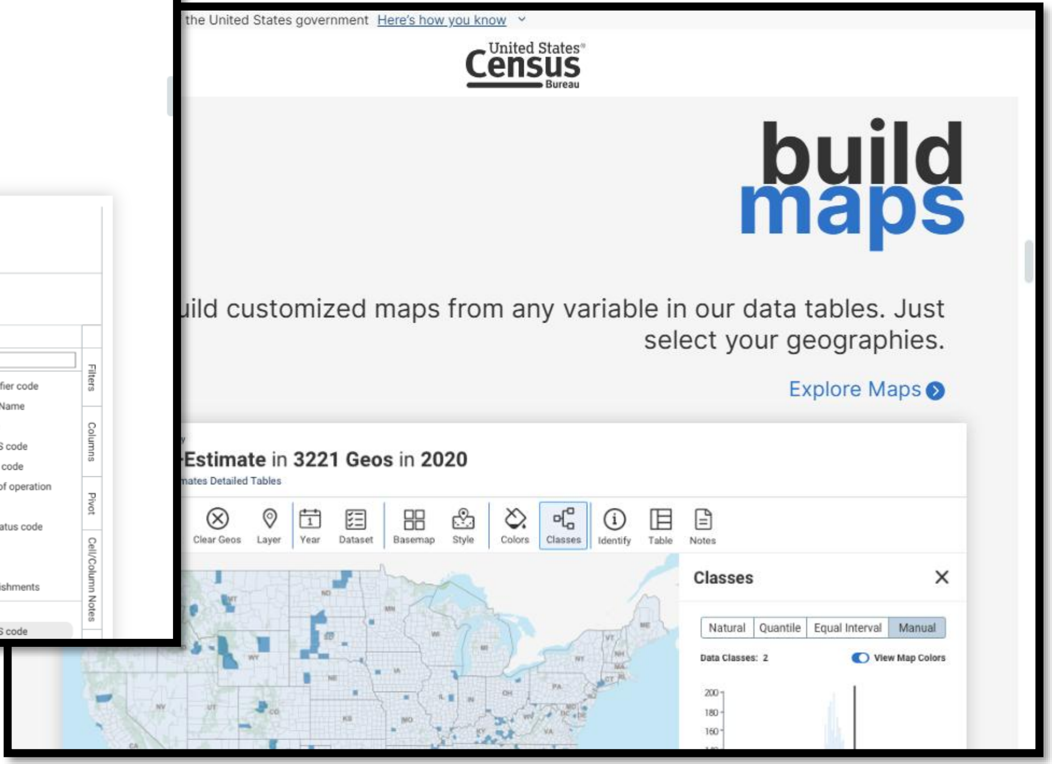
Institutional Datasets

Campus, city, or agency-specific data

Spreadsheets (CSV)

Tables with addresses or coordinates

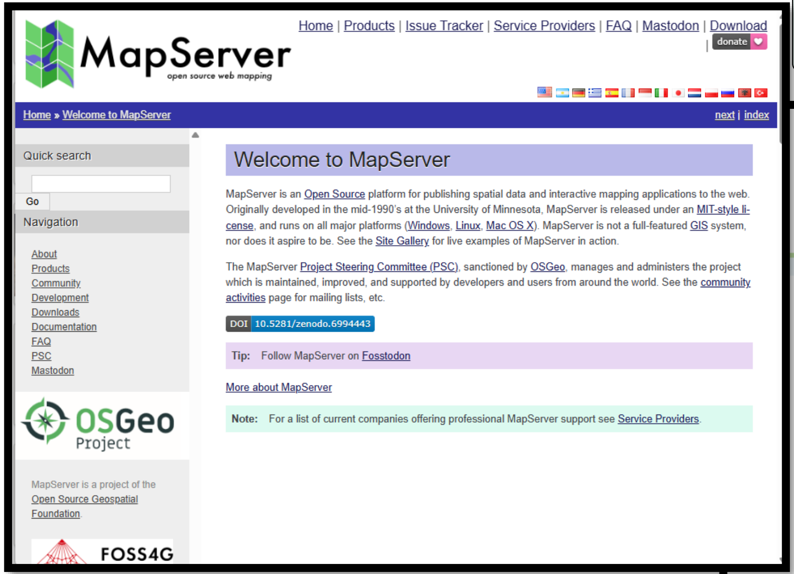




https://data.census.gov/

GIS Servers

Live data feeds from institutional systems



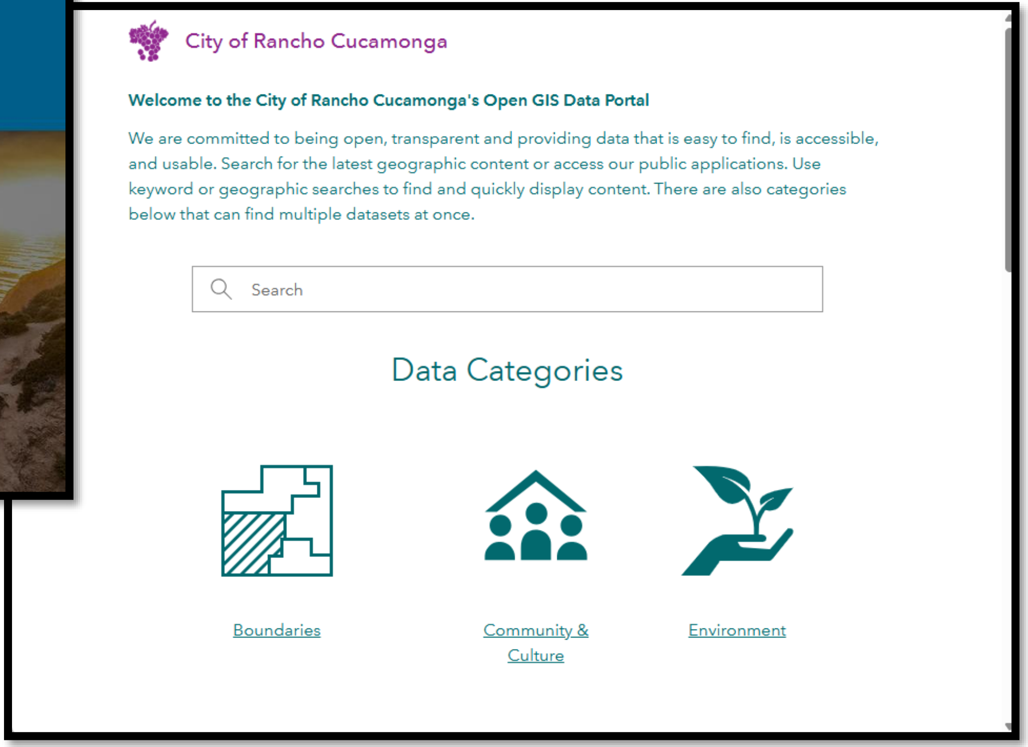
https://mapserver.org/

https://geoserver.org/

Open Data Portals

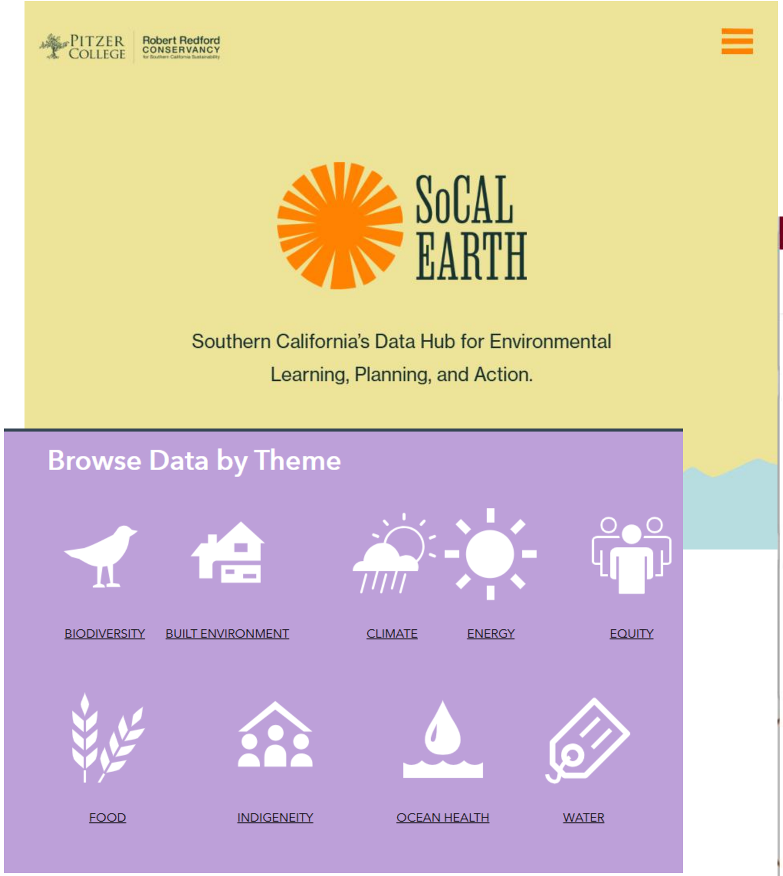
Public datasets from governments and organizations

https://data.ca.gov/

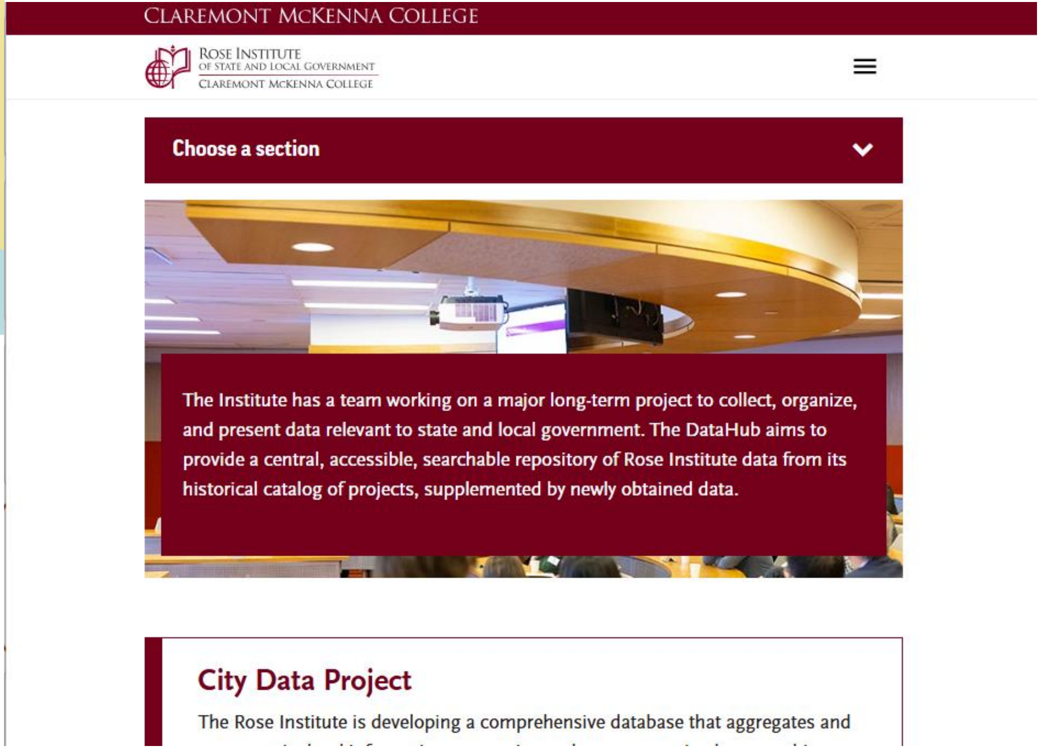


https://rcdata-regis.opendata.arcgis.com/

DATA HUBS



https://www.socalearth.org/



https://roseinstitute.cmc.edu/research/datahub

Institutional Datasets

Campus, city, or agency-specific data

THE CLAREMONT COLLEGES LIBRARY

Geographic Information System (GIS) Services

Dron...

Drone Flights

Request a Drone Consultation

The Digital Scholarship Unit at the Claremont Colleges Library offers consultation and support for faculty and students working with Unmanned Aerial Vehicles (UAV) or drone applications in GIS for research, teaching, and learning.

The library owns a DJI Phantom 4 Advanced and a Mavic 3E UAVs which we can use in support of research and teaching at the colleges (we do not loan the drones, but will set up a meeting with our drone pilot and the drone for your project).

Please get in touch with us if you need:

- The supervision/support of an FAA-licensed drone operator
- Use of photogrammetry applications (e.g., Pix4D) for processing drone imagery
- Consultations for support of student coursework involving drones
- Information about the Claremont Colleges drone policy and flight registration requirements

https://library.claremont.edu/gis/

CAL POLY

MY LIBRARY ACCOUNT

Robert E. Kennedy Library

VISIT SERVICES RESEARCH SPECIAL COLLECTIONS AND ARCHIVES

ABOUT

Robert E. Kennedy Library / Research Guides / Data Sources & Repositories / GIS Datasets

Search this Guide | Search

- Data Sources Home
- General Data Sources
- Data Catalogs
- Agriculture
- Archaeology
- Art
- Astronomy
- Biology
- Business
- Chemistry
- Earth Science
- Economic

https://guides.lib.calpoly.edu/c.php?g=261997&p=1750196

Step 3: Choose a Basemap

A basemap is the foundation of your map — it provides geographic context so your data has somewhere to live.

🗺️ Streets

Roads, labels, and boundaries — ideal for navigation and urban data.

🛰️ Satellite

Real-world imagery — great for land use, environment, and context.

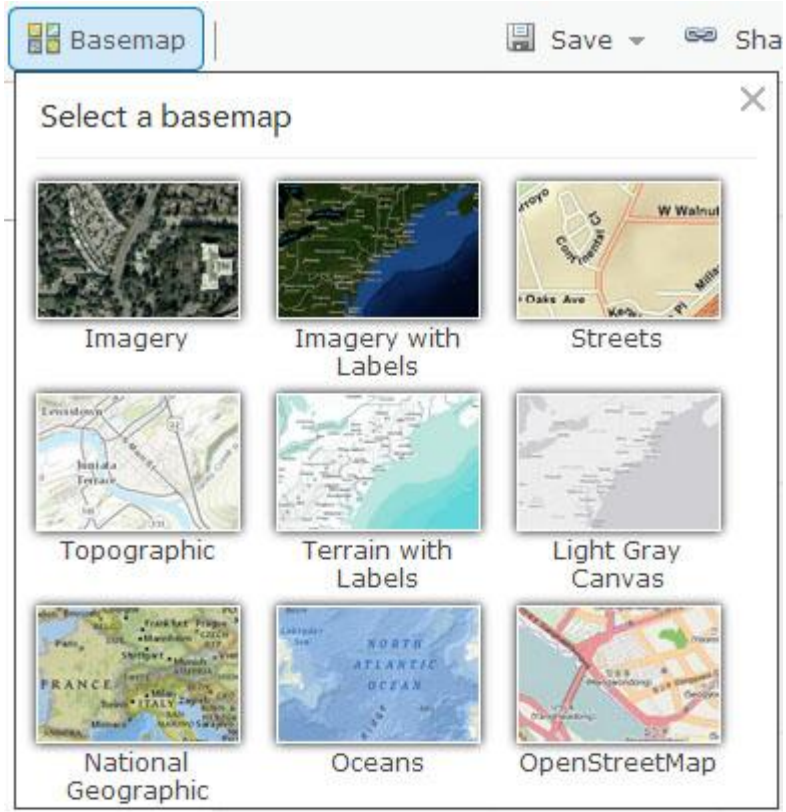
⛰️ Terrain

Elevation and topography — useful for natural features and landscapes.

🟣 Canvas

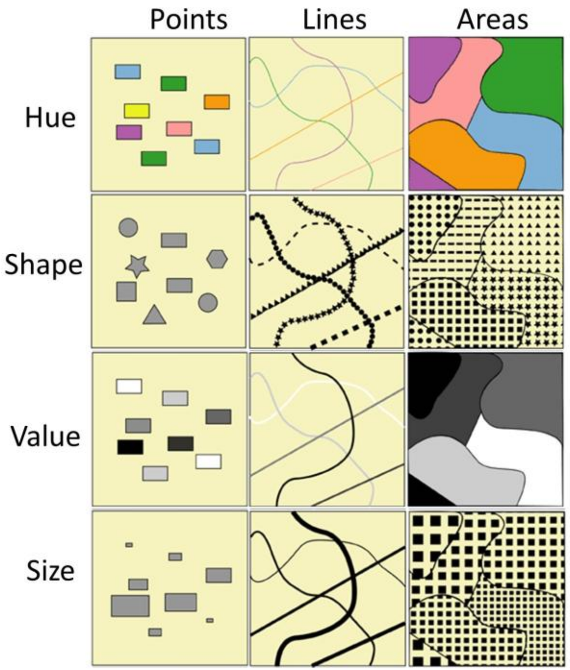
Minimal light or dark backgrounds — lets your data take center stage.

ⓘ **Why it matters:** The right basemap helps your audience immediately understand where they are and what your data means in context.



Step 4: Style Your Data

Styling turns raw data into a visual story. Thoughtful design choices help your audience understand patterns at a glance.

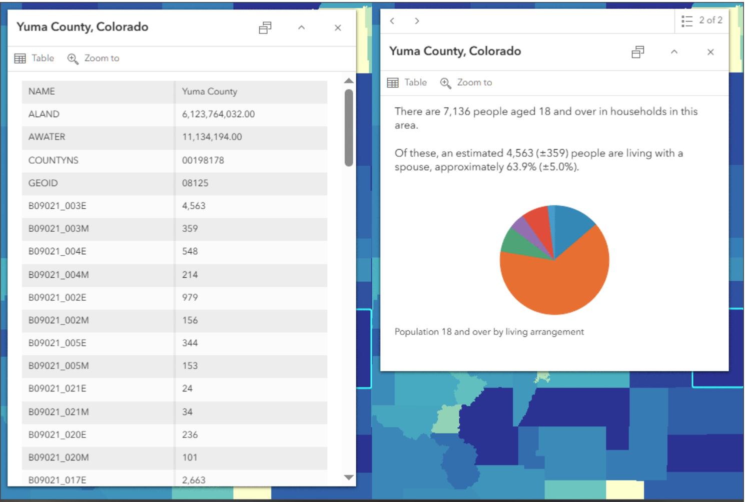


Design with Purpose

Every styling choice should serve your message:

- **Color** — distinguish categories or show value ranges
- **Size** — communicate magnitude or importance
- **Shape** — differentiate types of features

Goal: Make patterns easy to understand without reading a legend.



https://www.esri.com/arcgis-blog/products/arcgis-online/mapping/configure-pop-ups-basics

Step 5: Add Meaning with Pop-ups & Labels

Maps become truly useful when people can explore them. Pop-ups and labels turn a static image into an interactive experience.

💭 Pop-ups

Users click a feature to reveal details — names, values, descriptions, or links. Pop-ups make your map exploratory and informative.

🏷️ Labels

Key information displayed directly on the map — city names, site identifiers, or important features visible at all times.

What You've Built So Far

You now have the core building blocks of a functional GIS map. Here's a quick recap of your progress:

01

Basemap Selected

A geographic foundation that gives your data context and orientation.

02

Data Layers Added

One or more layers representing real-world features or phenomena.

03

Basic Styling Applied

Colors, sizes, or shapes that communicate meaning visually.

04

Spatial Understanding

A foundational grasp of how geographic data is structured and displayed.

Now It's Your Turn

Time to move from theory to practice. In this hands-on session, you'll build a real map from scratch.

1. **Open Your Tool**
   Launch the GIS mapping platform.
2. **Load Real Data**
   Import a dataset relevant to your campus or community.
3. **Build & Style**
   Apply basemaps, layers, and styling step-by-step.
4. **Explore & Ask**
   Use spatial thinking to answer real questions with your map.

Your facilitator **Daniel** will now guide you through the practical exercise. Don't hesitate to ask questions!